



CSCE 633-700: Machine Learning

Support Vector Machines

Overview

- Lagrange Optimization
- Support Vector Machines
 - Linearly separable case
 - Non-separable case
 - Hinge Loss
 - Multi-class SVMs

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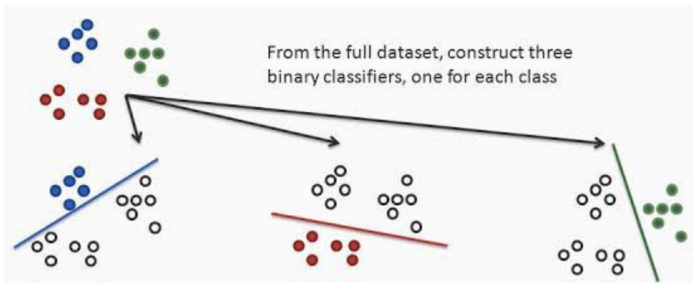
Multi-class kernel machines

One-vs-all approach

- Define K two-class problems, each separating one class from all others
- Learn K binary support vector machines $f_i(\mathbf{x})$, $i = 1, \dots, K$
 - Class 1: Examples from class i
 - Class -1: Examples from all classes besides class i , i.e. $\{1, \dots, K\} \setminus i$
- A sample point would be classified under a certain class if and only if that class's SVM accepted it and all other classes' SVMs rejected it

Multi-class kernel machines

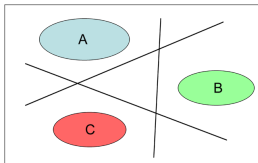
One-vs-all approach



Multi-class kernel machines

One-vs-all approach

When does one-vs-all fail?

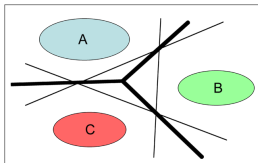


This method leaves regions of the space to be unaccounted for.

Multi-class kernel machines

One-vs-all approach

How to address regions of the space that are unaccounted for?



Decision during testing based on the class that has been selected with the highest confidence (e.g., highest distance to hyperplane).

Multi-class kernel machines

One-vs-one approach

- Define $K(K - 1)$ two-class problems, each separating class i from class j
- Learn $K(K - 1)$ binary support vector machines $f_{ij}(\mathbf{x})$
 - Class 1: Examples from class i
 - Class -1: Examples from class j
 - Note that $f_{ij} = -f_{ji}$
- During testing, the class chosen by the maximal number of SVMs is selected
- Alternatively (e.g., in case of tie)

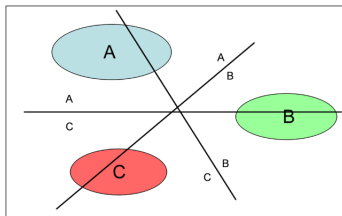
$$f(\mathbf{x}) = \arg \max_i \left(\sum_j f_{ij}(\mathbf{x}) \right)$$

(selects the class based on the sum of distance from all hyperplanes)

Multi-class kernel machines

One-vs-one approach

Example of decision boundaries



Multi-class kernel machines

Comparison of one-vs-all and one-vs-one approach

- One-vs-one
 - requires $O(K^2)$ classifiers instead of $O(K)$
 - but each classifier is on average smaller $O(2\frac{N}{K})$
- One-vs.-all approach solves $O(K)$ separate problems, each of size $O(N)$

Multi-class kernel machines

Multi-class formulation

- Define K weights for each class $\mathbf{w}_1, \dots, \mathbf{w}_K$ and K bias terms $w_{01}, \dots, w_{0K}$
- Training data $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, $y_n \in \{1, \dots, K\}$
- Optimization criterion

$$\min_{\mathbf{w}_1, \dots, \mathbf{w}_K} \frac{1}{2} \sum_{k=1}^K \|\mathbf{w}_k\|_2^2 + C \sum_{k=1}^K \sum_{n=1}^N \xi_{nk}$$

$$\text{s.t. } \mathbf{w}_{y_n}^T \mathbf{x}_n + w_{0y_n} \geq \mathbf{w}_k^T \mathbf{x}_n + w_{0k} + 2 - \xi_{nk}, \forall k \neq y_n$$

(i.e. so that the weight for each class yields a sufficient margin from the other classes)

What have we learnt so far

- SVM aims at finding the hyperplane from which instances have a margin of distance
- Prime and dual problem formulation (Lagrange multipliers)
- Support vectors: instances closest to separating hyperplane
- Linearly separable case: maximize margin of separation between two classes
- Non-separable case: look for the hyperplane that yields the least error (soft error)
 - Prime: minimizes Lagrangian wrt the primal variables of the problem
 - Dual: maximizes Lagrangian wrt multipliers
- Readings: Alpaydin 13.1-13.3